

Oil less Circuit Breaker

Lecture-6

2- Oil less Circuit Breaker

Such a type of high-voltage circuit breaker **does not use oil as an arc-quenching medium**. Different types of arc-quenching media can be used instead of oil. The following are the types of oil-less CBs using various arc quenching media;

2-1- Air Circuit Breaker

2-2- SF6 Circuit Breaker

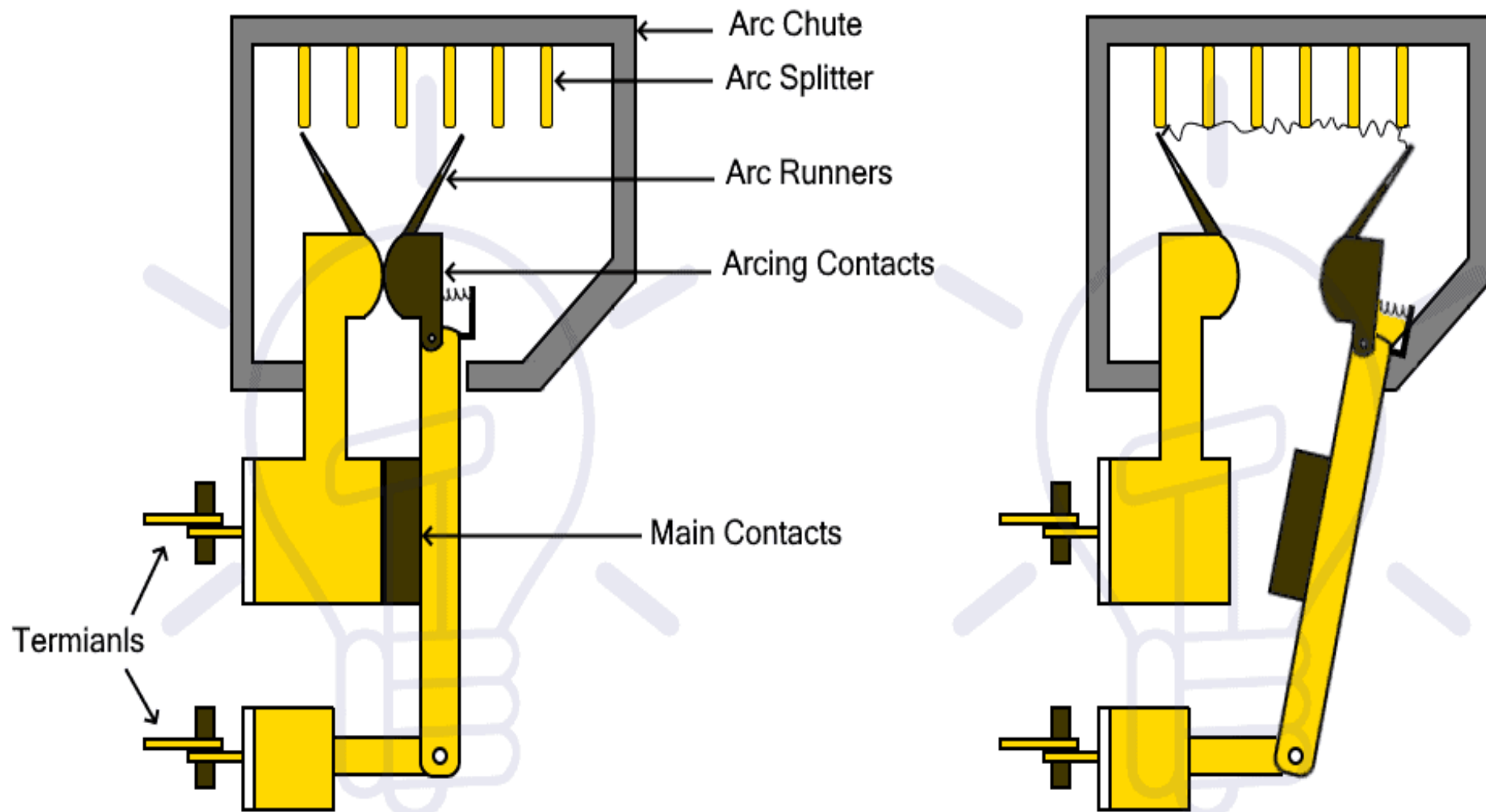
2-3- Carbon Dioxide Circuit Breaker

2-4- Vacuum Circuit Breaker

2-1 Air Circuit Breaker (ACB)

Air Circuit Breaker or (ACB) is a type of HV oil-less circuit breaker that **uses air as its arc extinguishing medium**. It is used for short circuit and overcurrent protection up to **15KV and 800 to 10K Amps**. It is preferable to oil circuit breaker due to the **absence of inflammable oil and the risk of fire hazards**.

The objective of a circuit breaker is to safely extinguish the arc and prevention of arc re-striking. In order to extinguish the arc, we have to increase the arc voltage (**minimum voltage required to maintain the arc**). ACB uses the air as a media to extinguish the arc. Unlike the other medias, the air can be used in various ways to extinguish the arc by using various **methods such as cooling the arc, increasing the arc length, splitting the arc and air blasting, etc.**



Air Circuit Breaker

In ACB, there are two pairs of contacts i.e. main contacts (made of copper) and arcing contacts (made of carbon). Under normal conditions, the main contacts are used for the normal supply of current. Under fault current, the main contacts open while the arcing contacts remain closed.

As soon as the main contacts open up, the current will flow through the arcing contacts. At this point, there is no arc and the main contacts are safe. The arc generates once the arcing contact opens up. The arc sweeps upwards, resulting in cooling and increasing the length of the arc. Thus, the arc is extinguished at zero current.

ACB is divided into the following types.

2.1.1 Plain Air Circuit Breaker

2.1.2 Arc-Chute Circuit Breaker

2.1.3 Magnetic Blow Out Circuit Breaker

2.1.4 Air Blast Circuit Breaker

2.1.1 Plain Air Circuit Breaker

A plain air circuit breaker is the simplest air circuit breaker. It is also known as cross blast circuit breaker. It has a chamber surrounding the main contacts. This chamber is called arc chute. It is used for extinguishing the arc made from refractory material. It has multiple small compartments made from the separation of metallic plates.

The metallic separation inside the arc chute acts as an arc separator, and it divides the arc into small arcs that increase the voltage required to maintain the arc. The air also sweeps the arc upwards, causing it to cool off. Thus arc is extinguished at zero current. It is used for low-voltage applications.

2.1.2 Air Chute Air Break Circuit Breaker

The air chute air break circuit breaker has two types of contacts i.e. main contacts and arcing contacts. The main contacts are made from silver-plated copper to reduce its resistance. The arcing contacts are made from a copper alloy that has very high heat resistance to absorb the damage from arcing.

Under normal operation, both contacts are closed. The main contacts conduct current due to their low resistance. When the main contacts open, the current diverts through the arcing contacts. Afterward, the arcing contacts are open where the arc is established and extinguished. The arcing contacts are easily replaceable in case of wear and tear.

2.1.3 Magnetic Blowout Air Circuit Breaker

Such type of Air circuit breaker has a blowout coil that generates a magnetic field. This magnetic field deflects the arc into the arc chute to extend its length as well as cool it. The arc is extinguished inside the breaker. The magnetic field does not directly extinguish the arc but merely deflects it, which is later extinguished by the air. It offers control over the arc to increase its voltage. These circuit breakers are used for up to 11KV.

2.1.4 Air Blast Circuit Breaker

Air Blast Circuit Breaker or (ABCB) uses a blast of compressed air for the arc interruption. The air is stored and compressed in a tank. This air is released through a nozzle at a very high speed to extinguish the arc.

They have a high voltage capacity of up to 450KV.

The air blast circuit breakers are divided into four types

2.1.4.1 Axial Blast Air Circuit Breaker

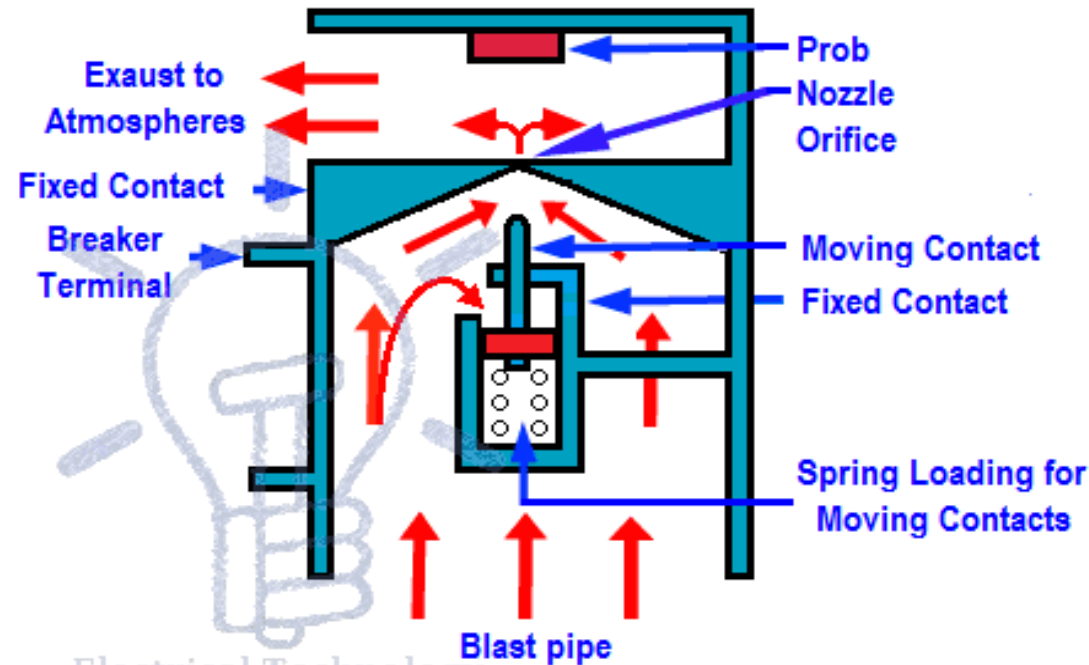
2.1.4.2 Axial Blast with sliding moving contact ACB

2.1.4.3 Radial Blast Air Circuit Breaker

2.1.4.4 Cross Blast Air Circuit Breaker

2.1.4.1 Axial Blast ACB

In axial blast ACB, the air blast **flows axially in the same direction as the arc**. The air blast lengthens and cools the arc at the same time and also increases the dielectric strength of the medium to prevent the arc from re-striking.



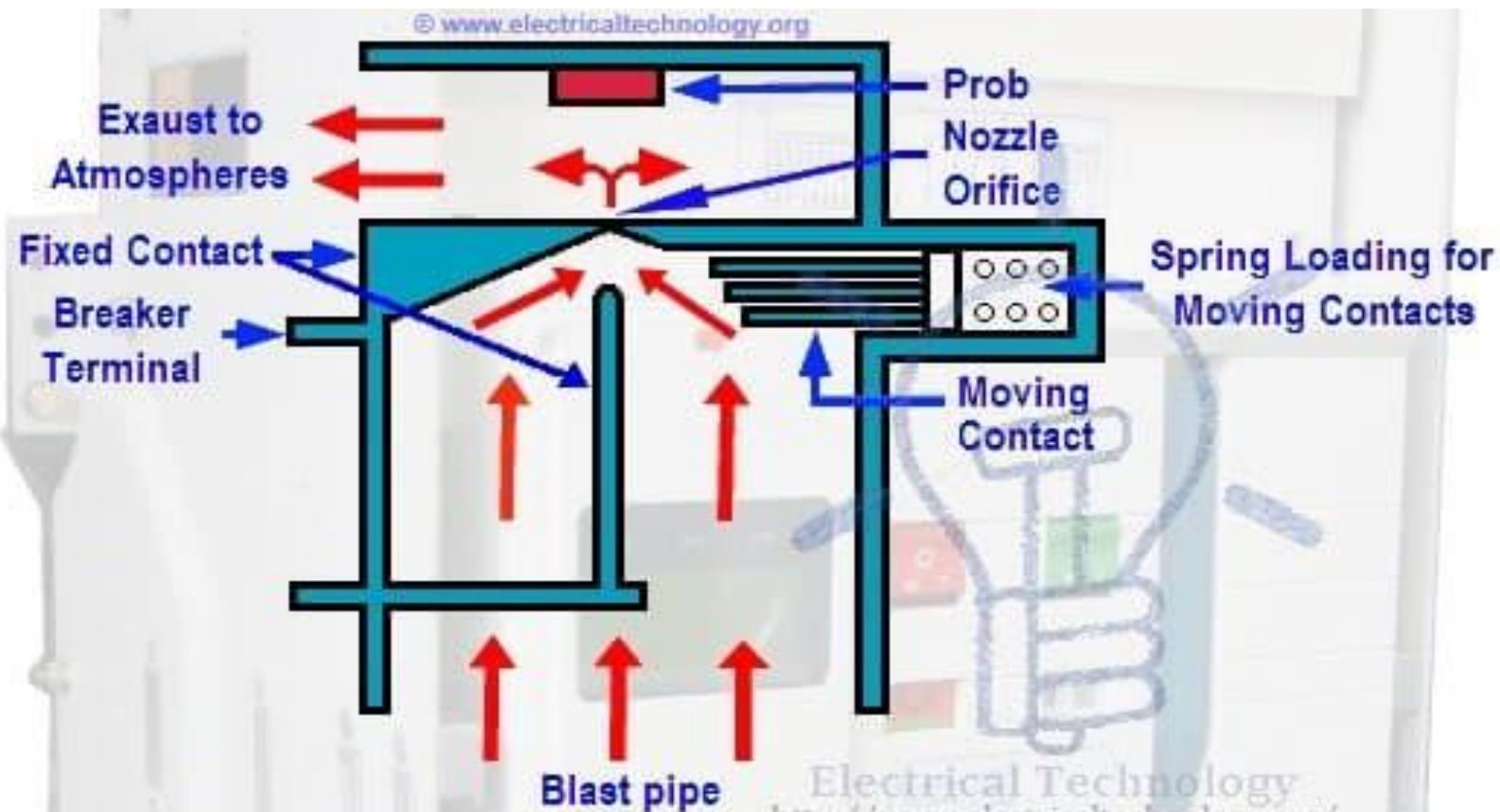
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**Schematic Diagram of
Axial Blast Air Circuit Breaker**

The moving contact under the force of spring is in closed position with the fixed contact. The fixed contact has a nozzle orifice blocked by the tip of the moving contact. The compressed air is stored in the tank below it. Upon fault condition, the compressed air is released that pushes the spring to pull the moving contact, thus opening the nozzle orifice in the fixed contact. The arc strikes between the contacts that is lengthened and cooled by the blast of air flowing through the orifice. It all happens simultaneously and the arc is extinguished.

2.1.4.2 Axial Blast ACB with Sliding Moving Contact

This type of breaker is a **modified form of the axial blast ACB**. It has a **moving contact horizontally fitted on a spring with a piston**. The fixed contact has the same kind of nozzle orifice, which is blocked by this moving contact. it has the same arc quenching operation as discussed in the previous ACB.

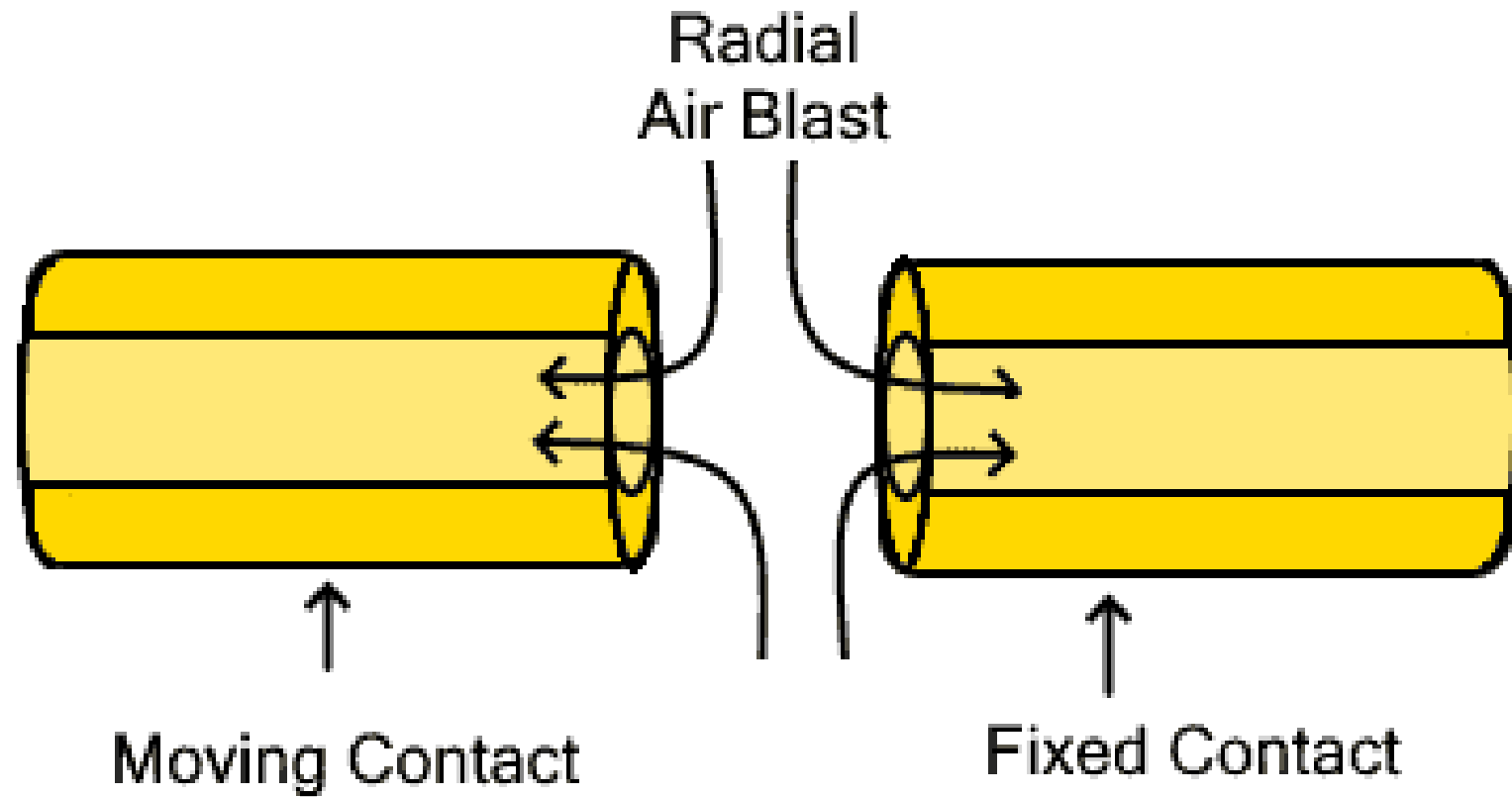


**Axial Blast Air Circuit Breaker
with Side Moving Contact**

2.1.4.3 Radial Blast ACB

In radial blast ACB, the contacts are hollow like a tube, there is free space inside the moving contact as well as the fixed contact. The free space is used for the flow of compressed air to cool the arc. The air blast flows radially inside the contacts along the arc; therefore, it is called radial blast ACB.

Under fault conditions, the arc is formed between the contacts when they are separated. The radial blast of air cools off the arc and increases the dielectric strength between the contacts. Upon zero current, the arc is extinguished.



Radial Blast ACB

2.1.4.4 Cross Blast ACB

In cross blast ACB, the air blast is directed at a right angle to the arc. The air blast is used to deflect and lengthen the arc into the arc chamber, where arc splitters split and lengthen the arc to extinguish it. There is exhaust in the arc chamber for the air to flow out.

Under fault conditions, the arc is formed between the contacts when they are separated. The radial blast of air cools off the arc and increases the dielectric strength between the contacts. Upon zero current, the arc is extinguished.

The air tank is fixed in a perpendicular direction to the movement of the contacts. Upon a fault condition, the contacts are opened and arc is established. At the same time, the air blast is released that sweeps the arc into the arc chamber. The arc splitters split the arc that is eventually quenched at zero current. The air blast also increases the dielectric strength of the medium between the contacts to prevent the arc from re-striking.

Advantages of ACB

- ACB does not have any fire hazards, unlike oil circuit breakers.
- ACB has a very high speed; its arc quenching is very fast.
- The speed of arc quenching is the same for all values of current.
- It is small due to the limited space required for an air blast to cool and quench the arc.
- It is very reliable and consistent because the short arc time cannot wear the contacts quickly.
- It requires less maintenance due to less wear of the contacts.
- ACB is fairly cheaper as air is used as the arc quenching medium.

Disadvantages of ACB

- . The air compressor needs to be maintained to have the correct pressure at all the time.
- . The air compressor takes up a large space.
- . The air pipes junction may leak air pressure.
- . There is a chance of a high rate rise of re-striking voltage and current chopping.
- . The air release produces a noise.
- . Air as compared to other insulating gases has lower extinguishing properties.